C3 Quantitative Chemistry – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Mass and Moles **Paper 1**

#	Question	Answer
1	Law of conservation of mass	Mass of products = mass of reactants
2	Relative formula mass (M_r)	Sum of the relative atomic masses in the formula
3	Why mass increases when a metal is heated in air	Oxygen from the air is added
4	Why mass decreases when a carbonate is heated	CO_2 gas escapes
5	Mole	Amount containing 6.02×10^{23} particles
6	Avogadro constant	6.02×10^{23} per mole
7	Equation for moles	Moles = mass ÷ M_r
8	Mass of 1 mole of a substance	Its M_r in grams
9	Limiting reactant	Reactant completely used up
10	Reactant in excess	Reactant left over

C3 Quantitative Chemistry – Knowledge Organiser

AQA GCSE Chemistry • Set 2: Concentration and Measurement **Paper 1**

#	Question	Answer
1	Concentration equation	Mass of solute ÷ volume of solution
2	Units of concentration	g/dm^3
3	1 dm^3 in cm^3	1000 cm^3
4	Convert cm^3 to dm^3	÷ 1000
5	Mass of solute from concentration	Concentration × volume
6	Numbers in a balanced equation show	Mole ratio
7	Mean	Total ÷ number of values
8	Range	Highest – lowest value
9	Uncertainty	Range of results about the mean
10	Effect of more solute in the same volume	Higher concentration

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AQA GCSE Chemistry • Set 3: Yield, Atom Economy, Titrations and Gases Paper 1 Triple only

#	Question	Answer
1	Percentage yield equation	Actual mass ÷ maximum theoretical mass × 100
2	Reasons yield is below 100%	Reversible reaction, losses when separating, side reactions
3	Atom economy equation	M_r of desired product ÷ total M_r of reactants × 100
4	Why high atom economy matters	Less waste; more sustainable
5	Concentration in mol/dm^3	Moles ÷ volume in dm^3
6	Glassware to measure 25.0 cm^3 of alkali	Pipette
7	Glassware to add acid in a titration	Burette
8	Volume of 1 mole of any gas at RTP	24 dm^3
9	Room temperature and pressure (RTP)	20 °C and 1 atm
10	Equation for gas volume at RTP	Moles × 24 dm^3